

MATH0107 Probabilistic Method in Combinatorics

<i>Year:</i>	2026–2027
<i>Code:</i>	MATH0107
<i>Level:</i>	7 (UG), 7 (PG)
<i>Normal student group(s):</i>	UG: Y4 Mathematics programmes PG: MSc Mathematics
<i>Value:</i>	15 credits (7.5 ECTS credits)
<i>Term:</i>	2
<i>Assessment:</i>	90% examination, 10% coursework
<i>Normal Pre-requisites:</i>	MATH0029 (Graph Theory and Combinatorics) or MATH0028 (Combinatorial Optimisation)
<i>Lecturer:</i>	Dr F Illingworth

Course Description and Objectives

The module introduces the probabilistic method, a powerful approach with many applications in combinatorics. The basic idea behind the method is that to prove that a combinatorial object with certain properties exists, it suffices to show that a random construction produces such an object with positive probability. The topics covered in this course will be presented along with applications in several areas in combinatorics, number theory and geometry.

Recommended Texts

- N Alon and JH Spencer, *The Probabilistic Method* (Wiley)

Detailed Syllabus

- **The basic method.** Ramsey numbers, domination, colouring hypergraphs, set systems.
- **Linearity of expectation.** Maximum cut, sum free sets, Turán’s theorem.
- **Alterations.** Large girth and chromatic number, colouring hypergraphs, dependent random choice.
- **The second moment.** Chebyshev’s inequality, distinct sums, number of prime divisors, random graphs.
- **Large deviations.** Chernoff bounds, consistent arcs in tournaments.
- **The local lemma.** Proof of the lemma, colouring hypergraphs, colouring real numbers, Ramsey numbers, finding directed cycles.
- Special topics may be drawn from the following, or other related topics:
 - **Correlation inequalities.** FKG inequality, intersecting families
 - **Martingales.** Azuma–Hoeffding inequality, chromatic number of random graphs, isoperimetric inequality in the hypercube
 - **Extras.** Independence number of triangle-free graphs, crossing numbers